

Quiz 3

✓ (1) A. $\cos \angle(a, b) = \frac{\text{pr}_a^\perp(b)}{|b|}$

B. $\text{pr}_a^\perp(b) = \text{pr}_a^\perp(b) \cdot \frac{a}{|a|}$

C. $\sin \angle(a, b) = \frac{\text{pr}_{\frac{a}{|a|}}^\perp(b)}{|b|}$

D. $\text{pr}_b^\perp(a) = \frac{\langle b, a \rangle}{\langle b, b \rangle} b$

✓ (2) $K = ?$ s.t. $d(P, l) = 5$.

$P(2, 3), l: 8x + 15y + K = 0$.

$$d(P, l) = \frac{|a \cdot x_0 + b \cdot y_0 + K|}{\sqrt{a^2 + b^2}} = \frac{|8 \cdot 2 + 15 \cdot 3 + K|}{\sqrt{8^2 + 15^2}} = 5$$

$\Rightarrow \dots \Rightarrow |61 + K| = 85$.

case I: $61 + K = 85 \Rightarrow K = 24$

case II: $61 + K = -85 \Rightarrow K = -146$.

Setup 1: $[A]_K = \begin{bmatrix} -5 \\ 2 \end{bmatrix}, [B]_K = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, [C]_K = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

✓ (3) $P \rightarrow$ orthogonal proj. of C on AB .

$\vec{AP} = \text{pr}_{AB}^\perp(AC) = \frac{\langle AC, AB \rangle}{\langle AB, AB \rangle} \cdot AB = \frac{9}{18} \cdot (3, 3) = \left(\frac{3}{2}, \frac{3}{2}\right)$

$\vec{AB} = (3, 3); \vec{AC} = (6, -3)$

$\langle AC, AB \rangle = \langle (6, -3), (3, 3) \rangle = 6 \cdot 3 + (-3) \cdot 3 = 9$.

$\langle AB, AB \rangle = (3 \cdot 3) + (3 \cdot 3) = 18$.

①

$$P = A + \vec{AP} = (-5, 2) + \left(\frac{3}{2}, \frac{3}{2}\right) = \left(-\frac{7}{2}, \frac{7}{2}\right).$$

✓ ✓ ④ $\rightarrow$ orthogonal proj. of $\vec{CB}$ on $\vec{CA}$.

$$\vec{CB} = (-3, 6)$$

$$\vec{CA} = (-6, 3).$$

$$n = \text{Proj}_{\vec{CA}}(\vec{CB}) = \frac{\langle \vec{CB}, \vec{CA} \rangle}{\langle \vec{CA}, \vec{CA} \rangle} \vec{CA} = \frac{36}{45} (-6, 3) = \left(-\frac{24}{5}, \frac{12}{5}\right).$$

$$\langle \vec{CB}, \vec{CA} \rangle = (-3) \cdot (-6) + 6 \cdot 3 = 36.$$

$$\langle \vec{CA}, \vec{CA} \rangle = (-6)^2 + 3^2 = 36 + 9 = 45.$$

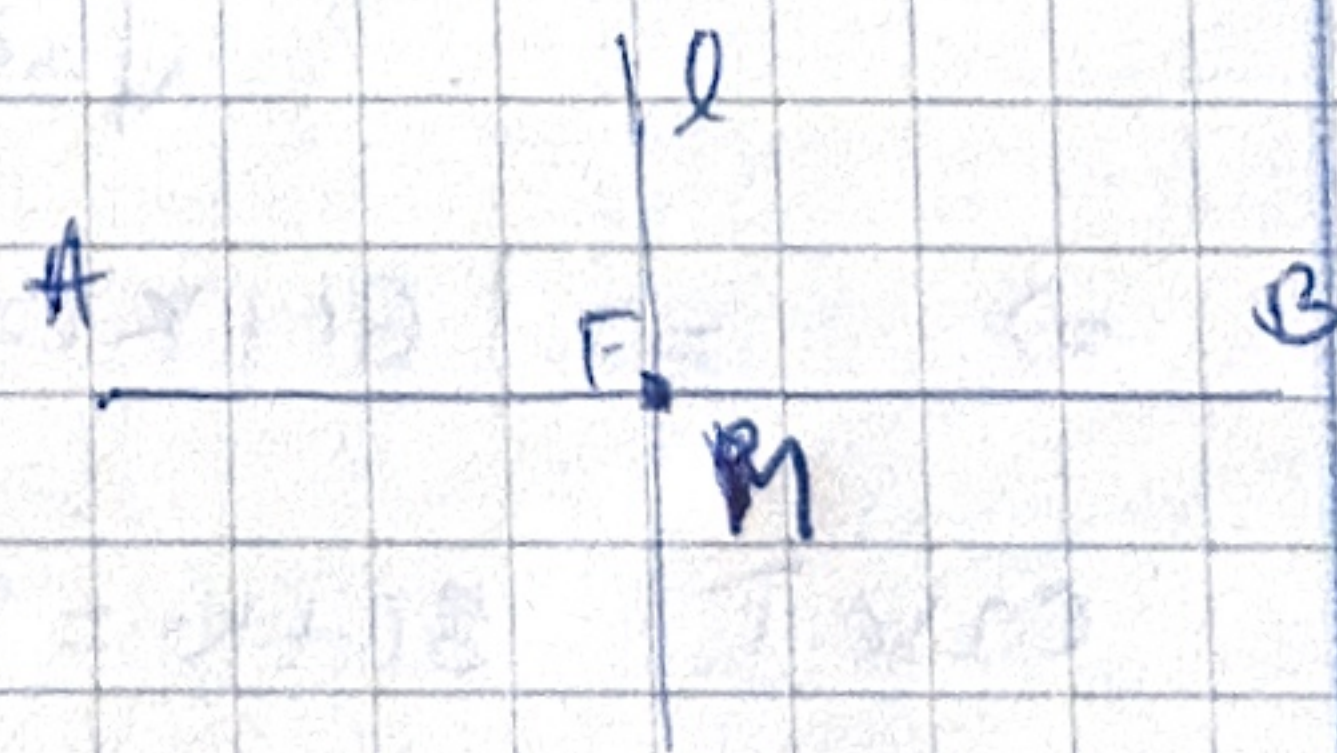
✓ ⑤ Scalar prod: $\boxed{A}$ bilinear.

$\boxed{C}$ pos. def.

$\boxed{D}$ symm. (not skew-symm.)

✓ ✓ ⑥ $l: 5x - 4y - 4 = 0.$

$$A(1, 0), B(-2, 1)$$



M - midpoint of AB.

$$M = \left(\frac{1+(-2)}{2}, \frac{0+1}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

$$\vec{AB} = B - A = (-3, 1) \text{ normal vector.}$$

• line passing through M with normal vector $n = (-3, 1)$

$$-3\left(x - \left(-\frac{1}{2}\right)\right) + 1\left(y - \frac{1}{2}\right) = 0.$$

$$\Rightarrow -3x - \frac{3}{2} + y - \frac{1}{2} = 0 \quad \left\{ \begin{array}{l} y = 3x + 2 \\ 5x - 4y - 4 = 0 \end{array} \right. \Rightarrow$$

$$\Rightarrow -3x + y - 2 = 0$$

$$\Rightarrow y = 3x + 2$$

②

$$2) 5x - 4(3x + 2) - 4 = 0$$

$$\Rightarrow -7x - 12 = 0 \Rightarrow x = -\frac{12}{7}$$

$$y = 3 \cdot -\frac{12}{7} + 2 = -\frac{22}{7}$$

$$\Rightarrow P\left(-\frac{12}{7}, -\frac{22}{7}\right)$$

✓

$$\textcircled{7} \quad g = 3i + j \Rightarrow g(3, 1, 0)$$

$$p = i + 2j + \lambda k \Rightarrow p(1, 2, \lambda)$$

$$\cos \alpha(p, g) = \frac{5}{12} = \frac{\langle p, g \rangle}{|p| \cdot |g|} = \frac{5}{\sqrt{10} \cdot \sqrt{5 + \lambda^2}} = \frac{5}{12}$$

$$\lambda = ?$$

$$\langle p, g \rangle = \langle (1, 2, \lambda), (3, 1, 0) \rangle = 1 \cdot 3 + 2 \cdot 1 + \lambda \cdot 0 = 5$$

$$|p| = \sqrt{1^2 + 2^2 + \lambda^2} = \sqrt{5 + \lambda^2}$$

$$|g| = \sqrt{3^2 + 1^2 + 0^2} = \sqrt{10}$$

$$\Rightarrow \sqrt{10} \cdot \sqrt{5 + \lambda^2} = 12 \quad |^2$$

$$\Rightarrow \sqrt{5 + \lambda^2} = 10(5 + \lambda^2) = 144 \dots$$

$$\Rightarrow \lambda^2 = \frac{94}{10}$$

$$\Rightarrow \lambda = \pm \sqrt{\frac{94}{10}}$$

✓

✓ $\textcircled{8}$. Distance between planes

$$\begin{cases} \pi_1: x - 2y - 2z + 7 = 0 \\ \pi_2: 2x - 4y - 4z + 17 = 0 \end{cases}$$

$$\pi_2: 2x - 4y - 4z + 17 = 0$$

$\bar{u}_1 \parallel \bar{u}_2$ if their normal vectors are proportional.

$$\bar{u}_1: x - 2y - 2z + 7 = 0 \Rightarrow m_1 (1, -2, -2)$$

$$\bar{u}_2: \frac{2x}{2} - \frac{4y}{2} - \frac{4z}{2} + \frac{17}{2} = 0$$

$$\Rightarrow x - 2y - 2z + \frac{17}{2} = 0.$$

$$\Rightarrow m_2 (1, -2, -2)$$

$$d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \frac{\left|7 - \frac{17}{2}\right|}{\sqrt{1^2 + (-2)^2 + (-2)^2}} = \frac{\frac{3}{2}}{3} = \frac{3}{6} = \frac{1}{2}$$

$$a = 1, b = -2, c = -2, d_1 = 7, d_2 = \frac{17}{2}.$$

$$\checkmark \textcircled{9} \begin{cases} \bar{u}_1: x - \sqrt{2}y + z - 1 = 0 \Rightarrow m_1 (1, -\sqrt{2}, 1) \\ \bar{u}_2: x + \sqrt{2}y - z + 3 = 0 \Rightarrow m_2 (1, \sqrt{2}, -1) \end{cases}$$

$$\cos \theta = \frac{|\langle m_1, m_2 \rangle|}{|m_1| \cdot |m_2|} = \frac{|-2|}{2 \cdot 2} = \frac{1}{2}$$

$$\langle m_1, m_2 \rangle = 1 \cdot 1 - \sqrt{2} \cdot \sqrt{2} - 1 \cdot 1 = -2.$$

$$|m_1| = \sqrt{1^2 + (-\sqrt{2})^2 + 1^2} = \sqrt{2 + 2} = 2.$$

$$|m_2| = \sqrt{1^2 + \sqrt{2}^2 + (-1)^2} = 2.$$

$\checkmark \textcircled{10}$ $A(x_A, y_A) = ?$; $A \in \ell: \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{1} = t$ which lies at distance $\sqrt{3}$ from plane $x + y + z + 3 = 0$.

$$\ell: \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{1} = t$$

$$\Rightarrow \begin{cases} x = 2t + 1 \\ y = 3t \\ z = t - 1 \end{cases}$$

$$d(A, \bar{u}) = \sqrt{3}.$$

$$d = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} = \sqrt{3}.$$

$$a=1, b=1, c=1, d=3.$$

$$A(2t+1, 3t, t-1).$$

$$\Rightarrow \frac{|1 \cdot (2t+1) + 1 \cdot 3t + 1 \cdot (t-1) + 3|}{\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow |2t+1+3t+t-1+3| = 3.$$

$$\Rightarrow |6t+3| = 3.$$

$$\text{Case I: } 6t+3=3 \Rightarrow 6t=0 \Rightarrow t=0.$$

$$\text{Case II: } 6t+3=-3 \Rightarrow t=-1.$$

$$\underline{t=0} \begin{cases} x=1 \\ y=0 \\ z=-1 \end{cases} \Rightarrow A(1, 0, -1)$$

$$\underline{t=-1} \begin{cases} x=-1 \\ y=-3 \\ z=-2 \end{cases} \Rightarrow A(-1, -3, -2).$$

✓ (11) $N \perp a(4, -2, -3)$ and $b(0, 1, 3)$.

$$\angle(v, Ox) \text{ acute, } |v|=26. \quad N=?$$

$$\Rightarrow v = a \times b = \begin{vmatrix} i & j & k \\ 4 & -2 & -3 \\ 0 & 1 & 3 \end{vmatrix} = \dots (-3, -12, 4).$$

(dir. vector)

$$\text{angle with the } x\text{-axis acute} \Rightarrow x > 0.$$

$$\underline{v} \cdot (-1) \Rightarrow \underline{(3, 12, -4)}.$$

✓

$$|v| = \sqrt{3^2 + 12^2 + (-4)^2} = \sqrt{169} = 13.$$

$$13 \rightarrow 26 \quad \Rightarrow \quad v = 2 \cdot (3, 12, -4) = (6, 24, -8)$$

(needed)